

Boosting Higher Alcohols Selectivity via Regulating Basicity of Ni/Hydroxyapatite in Ethanol Upgrading

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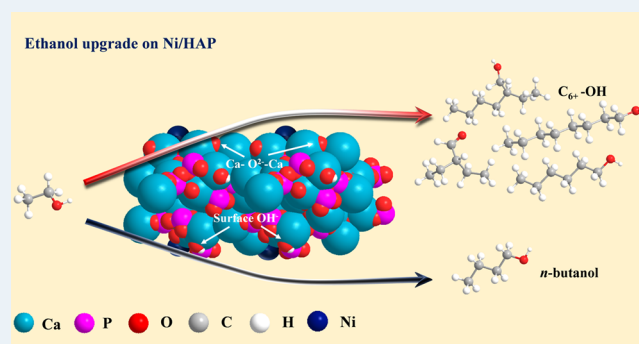
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ABSTRACT: Upgrading bioethanol to higher alcohols (C_{6+} -OH) is an environmentally friendly process for the production of high-value-added chemicals. However, a low selectivity for C_{6+} -OH is generally obtained due to the step-growth of the carbon chain, which leads to a preponderance of butanol in the product. In this work, we report base-Ni synergistic catalysis, with $Ni^{\delta+}$ on hydroxyapatite (HAP). The results showed that ethanol could be quickly dehydrogenated, continuously condensed, and finally hydrogenated to C_{6+} alcohols under the synergistic action of $Ni^{\delta+}$ and $Ca-[O^{2-}]-Ca$ active site pairs. The loading of Ni on the OH^- -rich HAP4 surface promotes the formation of more $Ni^{\delta+}$ and $Ca-[O^{2-}]-Ca$, which has been shown to be beneficial for ethanol upgrading. Especially, the ethanol reaction rate and the yield of C_{6+} -OH are directly proportional to the amount of $Ni^{\delta+}$ and the concentration of $Ca-[O^{2-}]-Ca$, respectively. The situ ethanol infrared results demonstrated the catalytic ability of different basic catalysts for intermediate aldehydes, which led to differences in the selectivity of C_{6+} -OH. A selectivity of 62.4% to C_{6+} -OH with an ethanol conversion of 63.6% has been achieved on HAP4-0.5Ni with abundant $Ni^{\delta+}$ and $Ca-[O^{2-}]-Ca$ sites under optimized conditions. Combined with the physical and chemical properties of the catalyst and reaction results, the mechanism of upgrading ethanol to C_{6+} -OH on Ni/HAP was also proposed.

KEYWORDS: bioethanol, higher alcohols, basic sites, Ni/hydroxyapatite, intermediate aldehydes



INTRODUCTION

The excessive consumption of carbon-based energy resources causes serious climate changes and corresponding environmental problems. It seems imperative to develop green and sustainable alternative fuels. At present, bioethanol has been used as a gasoline additive and is considered to be the most promising green fuel additive.¹ However, its hygroscopic properties and low energy density limit its further application as a diesel additive.² The higher alcohols, especially C_{6+} -OH, can avoid these disadvantages and are likely to be widely used as advanced fuel additives in the future.³ In addition, higher alcohols are widely used in plasticizers, cosmetics, fragrances, lubricants, and other fields. At present, the C_{6+} higher alcohols are mainly produced by the oxo process in industry, which is a complicated and energy-intensive process.⁴ The catalytic conversion of bioethanol to C_{6+} -OH will be a green route because of its simple process and the feedstocks derived from biomass.

Up to now, the catalysts used in ethanol upgrading include homogeneous ruthenium,^{5–7} iridium,^{8–10} and manganese complex catalysts,^{11,12} hydroxyapatite,^{13–16} metal oxides,^{17–22} and metal-supported heterogeneous catalysts with/without the assistance of liquid alkali.^{23–34} Generally, the upgrade of ethanol to higher alcohols was proposed to proceed by

“Guerbet” or “direct coupling” reaction pathways.^{35–37} Primary factors, including metal species, support, and promoters, are very important for the regulation of catalysts for performance improvement. The metal species (such as Ni,²⁸ Cu, and Pd³⁴) generally serve as the dehydrogenation/hydrogenation sites, and acid/base sites promote the aldol condensation. Typically, Mikkola and co-workers loaded Ru, Rh, Pd, Pt, Au, Ag, and Ni on Al_2O_3 for ethanol upgrading to study the effect of the active metal on the product selectivity and obtained >20% *n*-butanol yield, yet with a small amount of hexanol formed in the products.³⁸ The supports for the ethanol upgrading catalysts also include hydrotalcite, carbon materials, hydroxyapatite, and so forth.²⁵ Ueda and co-workers¹³ prepared HAP with various acid–base properties at different pH and showed different effects in the upgrading reaction of ethanol. Promoters like Zn, Sn, and S were found to be effective in the Ni-based catalysts for inhibiting the C–C

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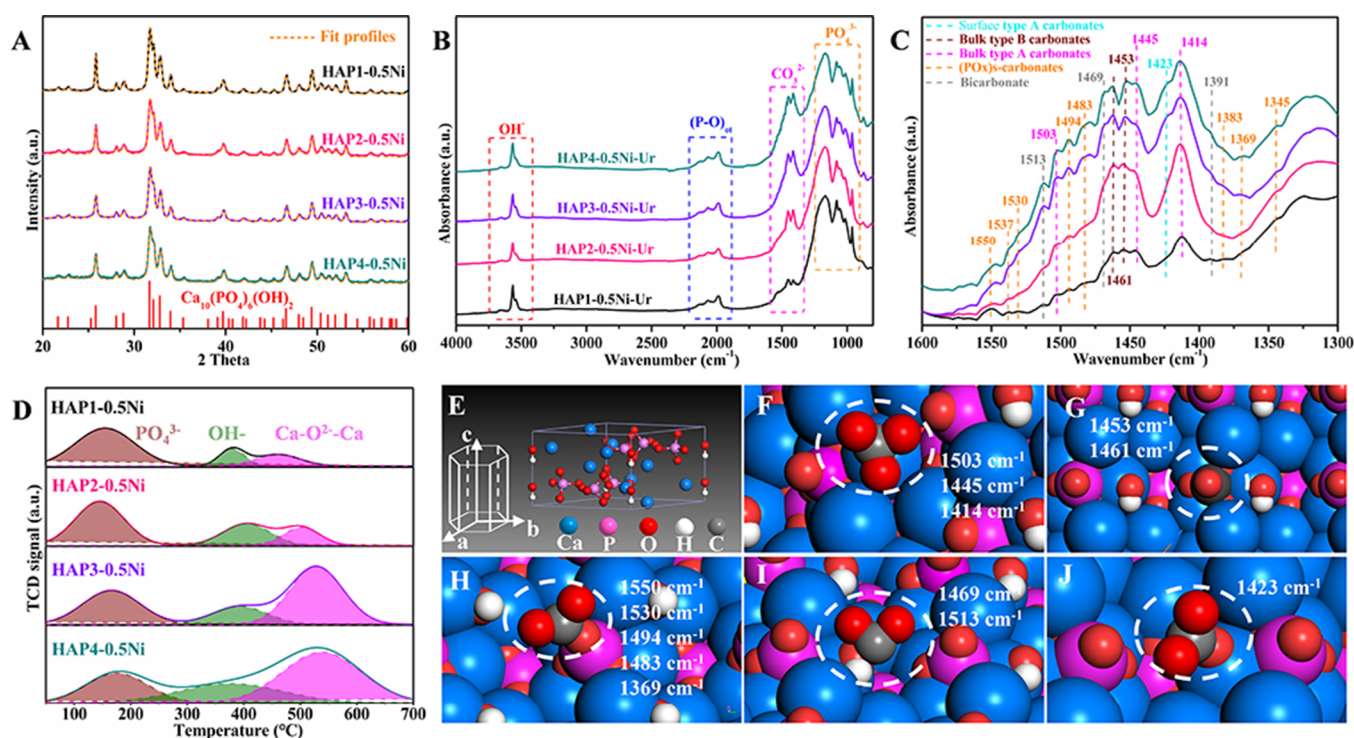


Figure 1. (A) Comparative XRD patterns, (B) DRIFTS spectra (4000–800 cm^{-1}), (C) DRIFTS spectra (1600–1300 cm^{-1}), and (D) CO_2 -TPD curves of $\text{HAP}_x\text{-0.5Ni}$ catalysts. (E) atomic structural representation and (F–J) CO_2 adsorption diagram of HAP. Note: $\text{HAP}_x\text{-0.5Ni-Ur}$ represents unreduced catalysts.

Table 1. Characteristic Properties of Catalysts^a

catalyst	pH^b	a ($\text{\AA}$) ^c	c ($\text{\AA}$) ^c	Ca/P molar ratio		Ni^d (wt %)	density of basic sites (mmol/g) ^f		
				bulk ^d	surface ^e		weak	medium	strong
HAP1–0.5Ni	7.5	9.425	6.872	um.	1.37	um.	0.236	0.040	0.037
HAP2–0.5Ni	8.5	9.420	6.874	1.63	1.41	0.56	0.216	0.124	0.069
HAP3–0.5Ni	9.4	9.417	6.880	um.	1.46	um.	0.212	0.128	0.396
HAP4–0.5Ni	10.2	9.416	6.883	1.73	1.48	0.54	0.208	0.143	0.445

^aum.: No measurement. ^bpH of precipitation suspension during catalyst synthesis. ^cData estimated by XRD. ^dData measured by ICP. ^eData estimated by XPS. ^fData measured by TPD.

cleavage and promoting C–C coupling to obtain longer alcohols in the ethanol upgrading reaction.^{39,40} The dispersion of the metal is also an important factor. He and co-workers demonstrated that the atomic Ru species play an important role in ethanol dehydrogenation and acetaldehyde condensation.⁴¹ Pang et al. synthesized the highly dispersed Ni-MgAlO catalysts and found that C_4 – C_8 alcohol selectivity reached 85% at 18.7% ethanol conversion over the optimal catalyst.²⁸

Inspired by the work of those who came before and refer to our previous work⁴² showing that the nickel-loaded biobased HAP catalyst achieved excellent catalytic effect, we chose Ni/HAP as the model material to investigate the precursor solution pH effect in the resulting supported Ni/HAP catalysts structure, physicochemical property, and catalytic effect. The results of this paper show that the precursor solution pH not only adjusted the base site distribution but also changed the interaction between the support and the Ni species. Different base sites were determined by in situ infrared and DFT calculations. The state of Ni species was studied through H_2 -TPR, XPS, and TEM. We found that $\text{Ni}^{\delta+}$ and $\text{Ca}[\text{O}^{2-}]_2\text{-Ca}$ site pairs play a key role in the boosting yield of $\text{C}_{6+}\text{-OH}$. The

mechanism of ethanol upgrading into $\text{C}_{6+}\text{-OH}$ on Ni/HAP was also discussed.

First, the model catalysts were prepared (for details, see the Supporting Information Section 1.2). Briefly, HAP was prepared by the coprecipitation method using $\text{Ca}(\text{NO}_3)_2$ and $(\text{NH}_4)_2\text{HPO}_4$ as precursors. The different HAP samples were obtained by controlling the pH of the precipitation solutions in the range of 7.5–11.2 using the aqueous ammonia solution and monitoring with a carefully calibrated pH meter (Mettler Toledo, SevenMult). In all of these conditions, white sediments can be obtained with a similar appearance. Then, an amount of 0.5 wt % nickel was loaded on HAP by a facile ethanol-assisted impregnation method followed by H_2 reduction. The reduced powders were labeled as $\text{HAP}_x\text{-0.5Ni}$ ($x = 1, 2, 3, 4$, and 5 when $\text{pH} = 7.5, 8.5, 9.5, 10.4$, and 11.2). Since the reaction results of HAP5-0.5Ni and HAP4-0.5Ni are similar (Table S1), the first four catalysts are used as typical catalysts for characterization.

RESULTS AND DISCUSSION

Structure and Physicochemical Properties of $\text{HAP}_x\text{-0.5Ni}$. The prepared catalysts have a similar appearance as

greyish powders. Figure 1A shows the XRD patterns of the four HAPx-0.5Ni catalysts. Cursorily glance, they are all similar in structure. Four patterns are all attributable to hydroxyapatite (PDF#74-0566) with a single phase. There is no oxidized or reduced nickel species diffraction peak in the XRD patterns, indicating a good dispersion of Ni components on the surface of the HAP supports.⁴³ Then, these four HAP structures were refined in the $p6_3/m$ space group, and the cell parameters are summarized in Table 1. The atomic structure of HAP in the stoichiometric form and the schematic representation of HAP crystal grown into a prism-like crystal are displayed in Figure 1E. The “ c ” values increase from 6.872 Å in HAP1-0.5Ni to 6.883 Å in HAP4-0.5Ni. This is because the Ca^{2+} ions gradually increased in the HAP structure with the increase of precipitation solution pH,¹³ as the results of the Ca/P molar ratio measured by XPS and ICP is shown in Table 1, and the Ca^{2+} has a larger radius than H^+ . Moreover, due to charge balance, the increase of Ca^{2+} will also result in the increase of OH^- which is an important source of HAP alkalinity.^{44,45}

The DRIFTS spectra for four samples are displayed in Figures 1B,C, S2, and S3. As shown in Figure 1B, the bands at 900–1200 cm^{-1} are attributed to the different vibration modes of P–O bonds in the PO_4^{3-} groups, and the bands at 1800–2200 cm^{-1} are assignable to the overtone and combination bands of PO_4^{3-} ions. However, the bands at 3500–3700 cm^{-1} represent the characteristic vibration modes for OH^- .⁴⁶ Similar to the XRD results, the DRIFTS results indicate the existence of a typical hydroxyapatite structure in the catalysts.

Moreover, in the hydroxyl vibration region shown in Figure S2, the bands of low intensity at 3600–3700 cm^{-1} are ascribed to different surface terminated PO–H groups,⁴⁷ and the main band at 3572 cm^{-1} is assigned to the stretching vibration of the lattice or surface OH^- ions. As shown in Figure 1C, the bands in the carbonate vibration region at 1369, 1483, 1494, 1530, and 1550 cm^{-1} corresponded to surface carbonates formed by the reaction of CO_2 with oxygen in PO_4^{3-} groups ((POx)_s-carbonate, Figure 1H), and the bands at 1469 and 1513 cm^{-1} are corresponded to surface bicarbonates formed by CO_2 interacting with surface OH^- (Figure 1I).^{48,49} However, the band at 1423 cm^{-1} is also a vibration band of surface carbonate (surface type A carbonate) formed by CO_2 interacting with surface $\text{Ca}[\text{O}^{2-}]\text{--Ca}$ which was formed by dehydroxylation after calcination at high temperature (Figure 1J).⁵⁰ The above different carbonate species indicate that there are three basic sites on the catalyst surface, which are in the order of O^{2-} in PO_4^{3-} , OH^- , and $\text{Ca}[\text{O}^{2-}]\text{--Ca}$, respectively. Figure S3 shows the DRIFTS spectra of the HAP4-0.5Ni catalyst before and after reduction. The disappearance of these carbonate species shows that the catalyst surface has excellent alkaline properties after reduction. In addition, the bands at 1453, 1461, and 1414, 1445, and 1503 cm^{-1} can be attributed to CO_3^{2-} substituting PO_4^{3-} (Figure 1G) and OH^- (Figure 1F), which indicates the formation of B-type and A-type apatite.⁵¹

The specific density and strength distribution of base sites were measured by CO_2 -TPD. Figure 1D shows the CO_2 -TPD curves of the four catalysts, and the strength and density of the basic sites are represented in Table 1. The evolution of CO_2 occurred in the 50–700 °C range, characteristic of catalysts containing base sites of different strengths. The desorption profiles can be divided into three peaks related to weak (approximately 50–287 °C), medium (287–438 °C), and strong (438–689 °C) base sites, attributed to O^{2-} in PO_4^{3-} , surface OH^- and surface $\text{Ca}[\text{O}^{2-}]\text{--Ca}$ species, as evidenced

by the DRIFT results.^{48,51} In order to further demonstrate the strength of these three basic sites, we undertook DFT calculations to investigate the adsorption of CO_2 over a HAP surface (Figure 2). The adsorption energy of CO_2 on PO_4^{3-} ,

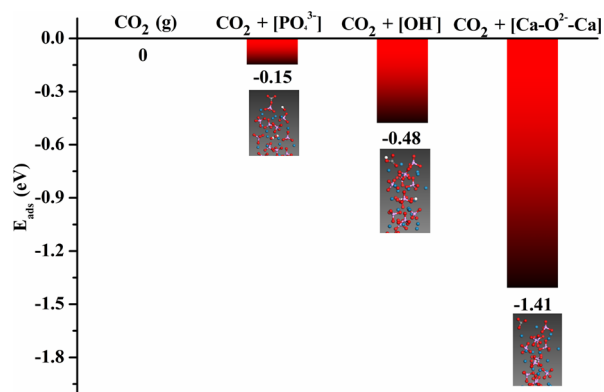


Figure 2. DFT calculation of CO_2 adsorption. (Ca: blue, P: pink, O: red, C: black, and H: white).

OH^- , and $\text{Ca}[\text{O}^{2-}]\text{--Ca}$ is -0.15 , -0.48 , and -1.41 eV respectively, which suggests that the adsorption of CO_2 by $\text{Ca}[\text{O}^{2-}]\text{--Ca}$ is the strongest, followed by OH^- , and the weakest by PO_4^{3-} . As shown in Table 1, there was no significant difference in the number of weak basic sites on the surface of the four catalysts. However, the amount of medium and strong base sites increases in the following order: HAP1-0.5Ni < HAP2-0.5Ni < HAP3-0.5Ni < HAP4-0.5Ni. In particular, the density of strong basic sites increased from 0.037 mmol/g in HAP1-0.5Ni to 0.445 mmol/g in HAP4-0.5Ni, which is attributed to the increase in the quantity of surface $\text{Ca}[\text{O}^{2-}]\text{--Ca}$ species.

Figure 3A shows the H_2 -TPR curves of the four catalysts. The diagrams of all catalysts are fitted to three peaks labeled I, II, and III. The peak I centered at 403 °C is associated with the reduction of Ni^{2+} to $\text{Ni}^{\delta+}$ ($0 < \delta < 2$), while the peak III centered at 467 °C is reduced to Ni^0 . The peak III above 600 °C is assigned to the reduction of a fraction of Ni^{2+} in the HAP structure.⁵² According to the peak area, the Ni^{2+} ions are more likely to incorporate HAP1 and HAP2 because of the cation vacancy left by Ca deletion. As the pH of the precipitated solution increases, the Ca content of HAP increases and it becomes difficult for Ni to substitute Ca into the HAP structure. Therefore, peak III is very small in HAP3-0.5Ni and HAP4-0.5Ni. However, due to the increase of OH^- and $\text{Ca}[\text{O}^{2-}]\text{--Ca}$ on the surface, more Ni has electron transfer with the carrier surface, so the area of peak I which represents $\text{Ni}^{\delta+}$ is in the order of HAP4-0.5Ni > HAP3-0.5Ni > HAP2-0.5Ni > HAP1-0.5Ni.

Upon the integration of peaks I and II, the corresponding H_2 consumption at 500 °C reduction which is the reduction temperature of the preparation catalyst was estimated and is summarized in Figure 3B. Figure S4 shows a standard curve of H_2 consumption using pure NiO to ensure the accuracy of the calculation results. The actual content of Ni measured by ICP has little difference from the target value, which is near 0.5 wt % (Table 1). As shown in Figure 3B, the H_2 consumption of HAP3-0.5Ni and HAP4-0.5Ni catalysts is more than four times the theoretical value. We checked the H_2 -TPR results of the HAP sample without metal, and no significant reduction peak was found (Figure S5). We assume that besides Ni

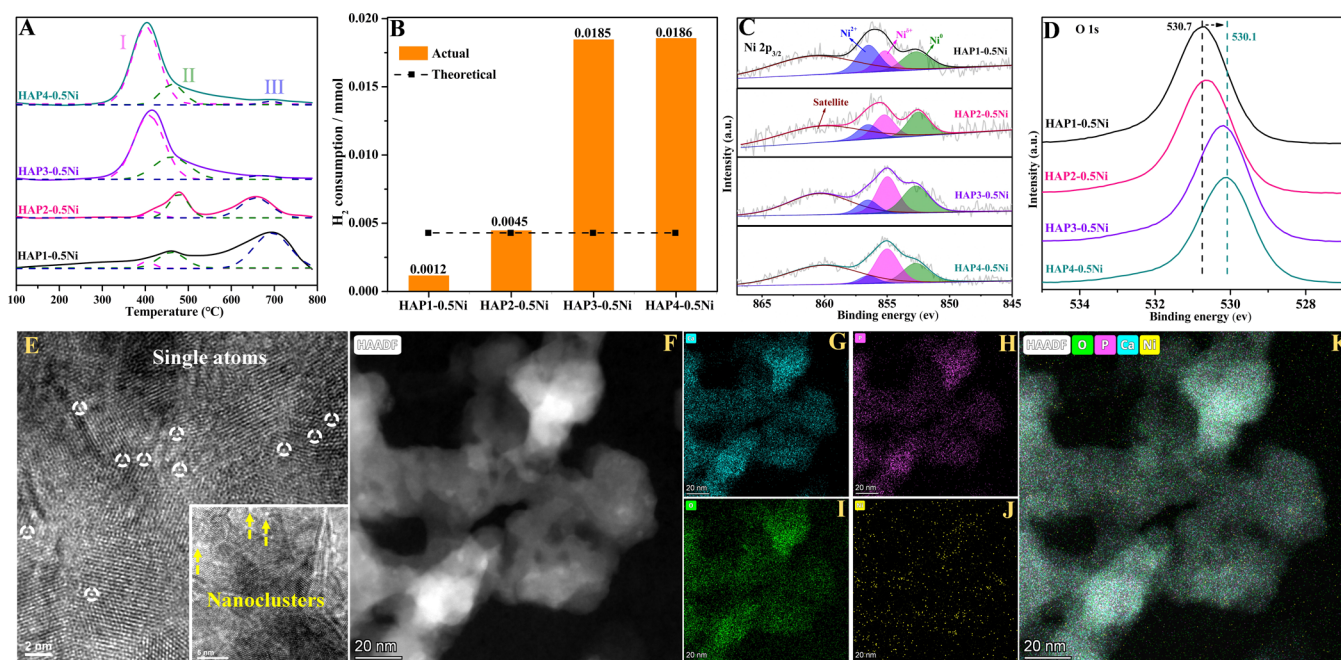


Figure 3. (A) H₂-TPR curves, (B) hydrogen consumption during temperature programmed reduction, (C) XPS Ni 2p_{3/2} spectra, and (D) XPS O 1s spectra of the HAPx-0.5Ni. (E) HRTEM image of HAP4-0.5Ni. (F) TEM image and (G–K) elemental mapping (Ca, P, O, and Ni) of HAP4-0.5Ni.

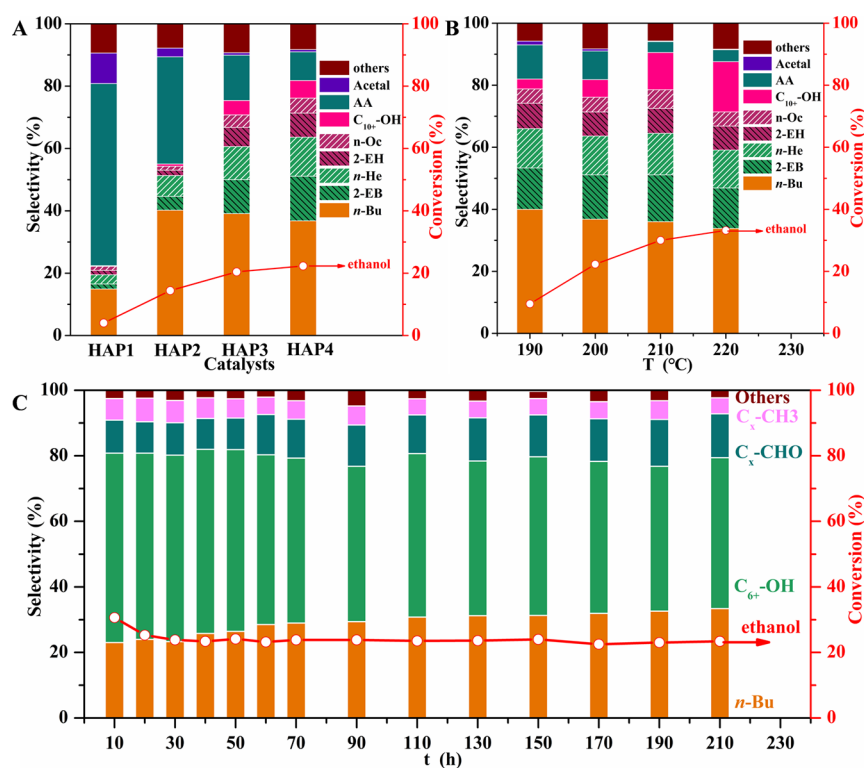


Figure 4. (A) Ethanol conversion and product distribution were measured on HAPx-0.5Ni at 200 °C. (B) Effect of the temperature on the reaction results on HAP4-0.5Ni. Conditions: 0.1 g catalyst, 1 g ethanol, 0.1 MPa N₂, 24h. Note, HAP1/2/3/4 refers to HAP1/2/3/4-0.5Ni, n-Bu: *n*-butanol, 2-EB: 2-ethylbutanol, n-He: *n*-hexanol, 2-EH: 2-ethylhexanol, n-Oc: *n*-octanol, AA: acetaldehyde. (C) Ethanol conversion over HAP4-0.5Ni catalyst. Conditions: 4% H₂/N₂ 2 MPa, 20 mL/min; 99.8% ethanol; 220 °C; WHSV = 0.7 h⁻¹.

reduction, the H₂ split occurred on the reduced Ni^{δ+} species and overflowed to the adjacent areas,⁵³ which indicates that Ni^{δ+} may have better dehydrogenation/dehydrogenation capacity than Ni⁰.

XPS analysis was utilized to disclose the existing state of the Ni species of reduced catalysts. As shown in Figure 3C, the Ni 2p_{3/2} spectrum shows four peaks at ca. 860.5, 856.2, 854.8, and 852.5 eV, which correspond to satellite peak, Ni²⁺ in the HAP structure, Ni^{δ+} species, and Ni⁰ species, respectively. With the

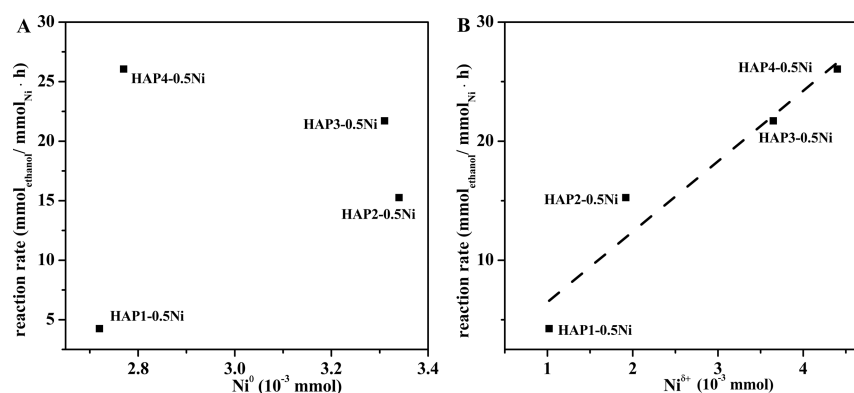


Figure 5. Correlation between the reaction rate of ethanol with the amount of (A) Ni⁰ and (B) Ni²⁺. Condition: 0.1 g of catalyst, 200 °C for 24 h, 0.1 MPa N₂, and 1 g of ethanol.

increase of the precipitated solution pH, the Ni²⁺ ions in the HAP structure decreased significantly, but the amount of Ni^{δ+} increased, which is consistent with the H₂-TPR results. More Ni species had electron transfer with the catalyst surface, thus forming more Ni^{δ+} species. As shown in Figure 3D, the binding energies of O 1s for the various HAPx-0.5Ni catalysts gradually move toward smaller values from HAP1-0.5Ni to HAP4-0.5Ni, also indicating the electron transfer from Ni to the HAP support.⁵⁴

The TEM images and elemental mapping of HAP4-0.5Ni are shown in Figure 3E–K. Figure 3E shows that HAP4-0.5Ni predominantly comprised atomically dispersed Ni, in addition to a small number of sub-1 nm Ni clusters. The elemental mapping of Ni shown in Figure 3J also proved that there is no obvious Ni agglomeration phenomenon. This is because the electron transfer between Ni and the HAP surface formed Ni^{δ+} species, making Ni highly dispersed, which allows it to catalyze the ethanol dehydrogenation and aldehyde hydrogenation reactions more efficiently.

Catalytic Conversion of Ethanol over HAPx-0.5Ni Catalysts. The HAPx-0.5Ni catalysts were used to upgrade ethanol to higher alcohols in a batch reactor. The catalytic activity and product distribution are shown in Figure 4A and Table S1. The liquid product yield calculated on a carbon basis on each catalyst was more than 97%, which mainly included acetaldehyde, butanal, *n*-butanol, *n*-hexanol, 2-ethyl butanol, *n*-octanol, 2-ethyl hexanol, C₁₀₊ alcohols, acetal, small amounts of alkanes, and other byproducts which are too few to quantify individually, as shown in Figure S6. A little of gaseous products are mainly hydrogen, methane, and carbon monoxide. Figure 4A shows that the ethanol conversion increased from 4.0% on HAP1-0.5Ni to 22.3% on HAP4-0.5Ni. The results of the nitrogen adsorption and desorption showed that the four catalysts had similar specific surface areas and pore volumes (Figure S1), which excluded their influence on the reaction. In general, the reduced metal will promote the dehydrogenation of ethanol. In fact, the ethanol barely converted on HAP4 without Ni loading (Table S1 Entry 10). We correlated the ethanol reaction rate with the amount of Ni⁰ and Ni^{δ+}, and the results are shown in Figure 5. The conversion of ethanol has no obvious relationship with Ni⁰, but it shows a positive relationship to the amount of Ni^{δ+}. As Ni^{δ+} increases from 1.0 to 4.4 μmol, the ethanol conversion rate per unit of Ni increases from 4.3 to 26.1 h⁻¹, showing a certain linear relationship ($R^2 = 0.9$). To further demonstrate that Ni^{δ+} plays a role that is more important than that of Ni⁰ in this reaction,

we loaded HAP4 with 5% Ni to increase the content of Ni⁰ and catalyze the conversion of ethanol. The results (Table S1 Entry 11) showed that the ethanol conversion was 23.6%, which was no better than HAP4-0.5Ni, indicating that Ni^{δ+} plays a major role in the dehydrogenation of ethanol. However, this result cannot prove that the role of Ni⁰ can be ruled out completely. Further studies will be conducted on that in future reports.

In addition, the total alcohol selectivity also increased on the four catalysts, and the selectivity of aldehyde and acetal which are the main byproducts showed a contrary trend. This must be due to the enhancement of the basicity of the catalysts, which can promote the aldol reaction of the aldehydes. The details of product distribution show that the selectivity of C₆-OH and C₈₊-OH increased significantly from 4.6 and 2.9% on HAP1-0.5Ni to 26.8 and 18.2% on HAP4-0.5Ni. In order to eliminate the effect of conversion, the conversions on HAP2-0.5Ni, HAP3-0.5Ni, and HAP4-0.5Ni were controlled at about 15%, the HAP4-0.5Ni still showed its superiority in C₆₊-OH selectivity, as shown in Table S1 (entries 2, 5, and 6). The results show that the catalyst with strong alkaline properties has higher selectivity for C₆₊-OH, which is due to the promotion of the aldol reaction by strong base sites. The aldol reaction is the step of forming a C–C bond, so its activity has a great influence on the carbon chain length of the final product. Therefore, the selectivity of C₆₊-OH increased gradually with an increase in the basicity of the catalysts. In addition, we also synthesized the corresponding HAP5-0.5Ni in the case of pH 11.2 to catalyze the reaction. The ethanol conversion was 22.6%, and the selectivity of C₆₊-OH was 45.8%, which is similar to that of HAP4-0.5Ni, as shown in Table S1 (Entry 12). Therefore, the four catalysts presented are selected as typical catalysts.

The effect of the temperature on ethanol conversion was also investigated on HAP4-0.5Ni. As shown in Figure 4B and Table S1 (Entry 2, 7, 8 and 9), the ethanol conversion increased from 9.5 to 33.2% with the reaction temperature increasing from 190 to 220 °C. The selectivity for C₆₊-OH reached its highest value 54.6% at 210 °C. The improved conversion indicates that the increase in temperature promotes ethanol dehydrogenation. In addition, as the temperature increased from 190 to 210 °C, the selectivity of acetaldehyde decreased and the selectivity of total alcohols and C₆₊-OH increased, indicating that the temperature increase is also conducive to aldol condensation. However, upon further increasing the temperature to 220 °C, the selectivity for all

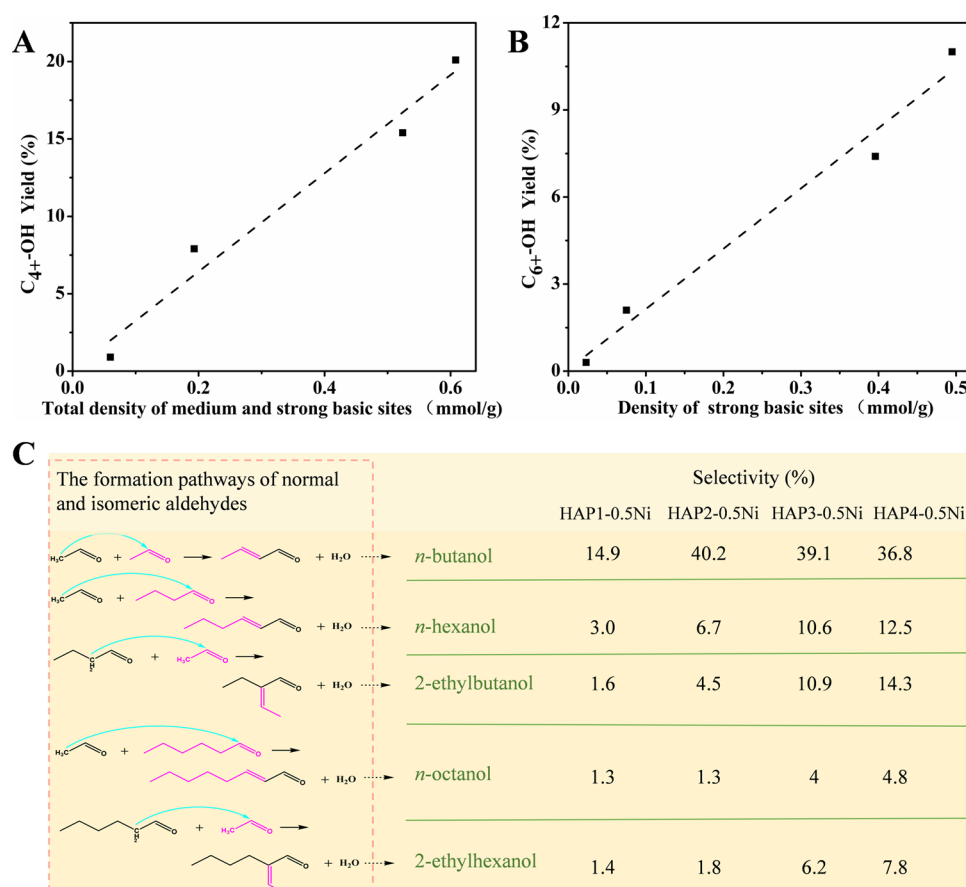


Figure 6. (A) C_{4+} -OH yield as a function of the total density of medium and strong basic sites. (B) C_{6+} -OH yield as a function of the density of strong basic sites. (C) Formation pathways of normal and isomeric aldehydes, in which deprotonation of species on the arrow end and subsequent nucleophilicity attach to the species on the arrowhead. Reaction conditions: 0.1 g catalyst, 200 °C, 24 h, 0.1 MPa N_2 , 1 g ethanol.

alcohols decreased due to the increase in alkanes and other byproducts. This is probably because the activity of aldehyde decarbonylation and self-condensation increased.

The influence of catalyst dosage on ethanol upgrading was investigated, and the result is shown in Table S1 (entries 2, 15, 16, and 17). With an increase in the catalyst amount, the ethanol conversion and C_{6+} -OH selectivity both increased. When the weight percentage of the catalyst was 20%, the ethanol conversion and C_{6+} -OH selectivity reached 40.0 and 62.8%, respectively. However, when the catalyst weight was further increased to 0.4 g, the ethanol conversion was only improved by 6%, and the C_{6+} -OH selectivity was almost unchanged, which is most likely because too much catalyst powder affects the contact between the reactants and the catalyst. By adding 2.5 mL of toluene as a solvent to avoid this problem, the conversion of ethanol reached 63.6%, and the selectivity of C_{6+} -OH remained above 62% (Table S1, Entry 18).

In order to further evaluate the performance of the catalyst and explore its industrial application prospects, HAP4-0.5Ni was also tested in a continuous-flow fixed-bed reactor. As shown in Figure 4C, after 10 h, the conversion of ethanol is stable at about 23%, and the selectivity of butanol and C_{6+} -OH is about 30 and 50%, respectively. The byproducts mainly include aldehydes, alkenes, and other condensation products. Moreover, the HAP4-0.5Ni catalyst exhibited superior stability over 200 h on-stream evaluation, showing a promising prospect for industrial applications.

Role of $Ni^{\delta+}$ and Basic Sites on the Production of C_{6+} Alcohols. The process of upgrading ethanol to higher alcohols through the Guerbet reaction usually involves dehydrogenation, aldol reaction, and hydrogenation. Among them, the C-C bond coupled by aldol reaction is the key step for carbon chain growth. Figure 6A shows the C_{4+} -OH yield as a function of the total density of medium and strong base sites, indicating that the base sites played a significant role in the total alcohol yield. Combined with the reaction results on HAP1-0.5Ni, the medium and strong base sites could promote the aldol reaction, while the weak base sites had almost no effect. Figure 6B shows that the yield of C_{6+} -OH increased almost linearly with the increase of strong base sites, indicating that the presence of strong base sites further enhanced the activity of the aldol reaction and promoted the growth of carbon chains, which could not be achieved by only relying on medium base sites. These results demonstrate the importance of strong base sites for the formation of C_{6+} -OH. In order to further determine the key role of base sites, CO_2 was used to poison the base sites on the HAP4-0.5Ni catalyst, and the catalytic performance was evaluated and is shown in Table S1 (Entry 12). After preabsorption of CO_2 , ethanol conversion and C_{6+} -OH selectivity decreased obviously, indicating the base sites played an important role. In addition, we mixed Rainey-Ni with HAP4 to catalyze ethanol upgrading; the selectivity of C_6 -OH was 18%, and C_{8+} -OH was only 2.2%. The result suggested that the presence of $Ni^{\delta+}$ is essential for the high selectivity of higher alcohols. The synergistic effect

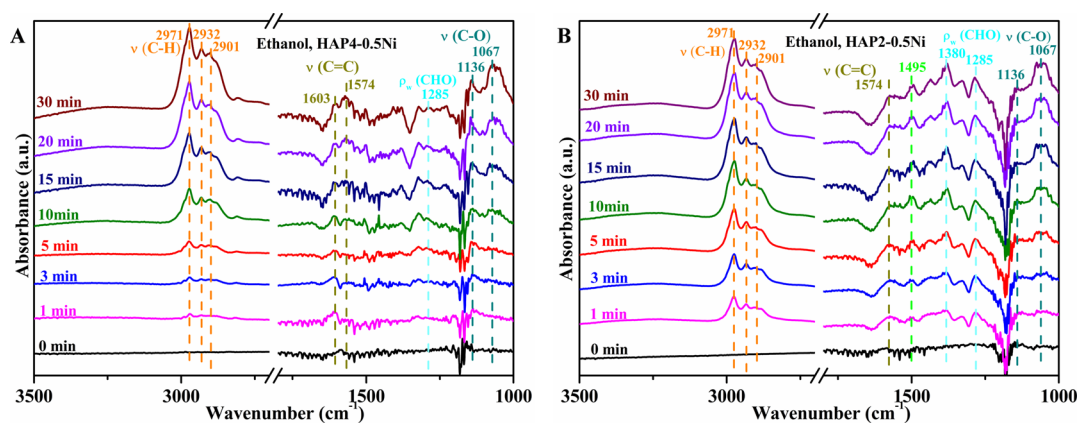


Figure 7. In situ DRIFT spectra were recorded over (A) HAP4-0.5Ni and (B) HAP2-0.5Ni at 200 °C. Reaction activity was measured in streamflow of 5.0 vol % C₂H₅OH/N₂.

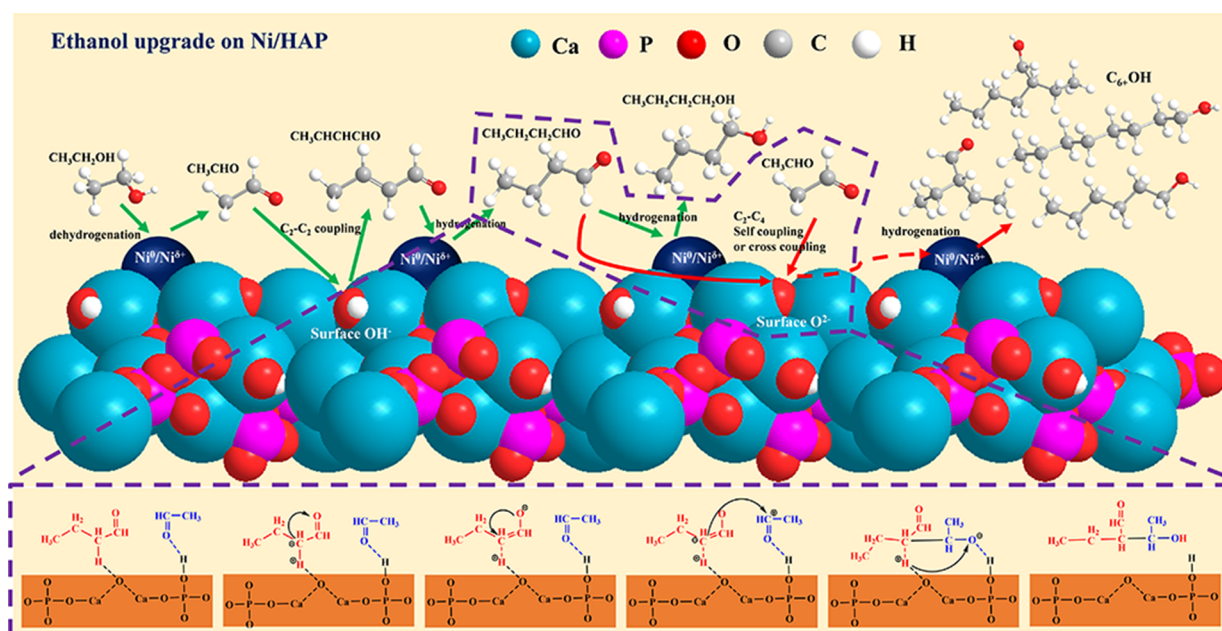


Figure 8. Proposed reaction network for ethanol conversion.

of Ni^{δ+} and Ca-[O²⁻]-Ca catalyzes the upgrading of ethanol to higher alcohols. In order to verify that the medium and strong basic sites are derived from the hydroxyl group, we prepared chloride-substituted chlorapatite and then used it for this reaction. As shown in Table S1 (Entry 13), ethanol conversion and C₆₊-OH selectivity also decreased obviously, indicating the key role of the hydroxyl group and its dehydrated Ca-[O²⁻]-Ca as medium and strong basic sites.

Moreover, the aldol reaction usually proceeded via a nucleophilic attack mechanism, in which the deprotonation of aldehydes is the rate-limiting step.⁵⁵ As shown in Scheme S1, the α-C atom loses H atoms to form enolates on the surface of the catalyst. The enolate is then nucleophilically added to the carbonyl group of another aldehyde to form a hydroxyl aldehyde. However, the α-H pK_a of the aldehydes will increase with the carbon chain growth due to the inductive effect, which causes the difficulty of deprotonation in the order acetaldehyde, butyraldehyde, and hexaldehyde. Figure 6C displays the formation pathways of normal and isomeric C₆ and C₈ aldehydes, which are also the mechanisms for the formation of isomeric and normal alcohols. The normal C₆ and

C₈ aldehydes are derived from the nucleophilic addition of deprotonated acetaldehyde to other aldehydes, while the isomerized aldehydes are derived from the addition of deprotonated C₄ or C₆ aldehydes to other aldehydes. Considering the difficulty of deprotonation of aldehydes with different carbon chain lengths, the isomerized products, aldehydes or alcohols, are more difficult to produce. Figure 6 shows that the selectivity of isomeric C₆ and C₈ alcohol increases from 1.6 and 1.4 on HAP1-0.5Ni to 14.3 and 7.8 on HAP4-0.5Ni, indicating that the higher basicity of the catalyst promoted deprotonation of the C₄₊ aldehydes. This is also the reason for the higher selectivity of C₆₊-OH with increasing basicity.

Mechanism of Ethanol Upgrading to Higher Alcohols on HAPx-0.5Ni. To further explore the C₆₊-OH formation mechanism on Ni/HAP, in situ DRIFT experiments were conducted on HAP2-0.5Ni and HAP4-0.5Ni (Figure 7). As shown in Figure 7A, the bands at 2971, 2932, and 2901 cm⁻¹ are assigned to ν(C-H) of ethanol and product alcohols, demonstrating the upgrading of ethanol to higher alcohols.^{14,33} The band at 1285 cm⁻¹ is assigned to ρ_w(CHO) of

acetaldehyde, and the bands at 1603 and 1574 cm^{-1} are attributed to the $\nu(\text{C}=\text{C})$ and $\delta(\text{C}=\text{C})$ modes of acetaldehyde and crotonaldehyde, respectively.⁵⁶ These bands are relatively weak and do not change much later, indicating that the intermediate aldehydes are quickly coupled to strong basic sites on HAP4–0.5Ni. Besides the characteristic peaks of alcohols, the characteristic peaks of acetaldehyde increased significantly on HAP2–0.5Ni (Figure 7B), indicating that there are not enough strong basic sites to promote acetaldehyde condensation and produce a large number of acetaldehyde byproducts. These results of DRIFTS further proved the positive effect of basic sites on the growth of carbon chains.

The possible reaction mechanism of ethanol upgrading to higher alcohols on Ni/HAP was proposed according to reaction and characterization results and is shown in Figure 8. Ethanol is first dehydrogenated at Ni sites to produce acetaldehyde. Then, acetaldehyde molecules undergo aldol condensation and are dehydrated to crotonaldehyde at the surface of OH^- or $\text{Ca}[\text{O}^{2-}]\text{Ca}$ (medium and strong base sites). Finally, crotonaldehyde is hydrogenated twice to butanol. With the increase of strong base sites, the generated acetaldehyde and butaldehyde can be further coupled with each other to produce long-chain aldehydes, followed by hydrogenation to produce higher alcohols. Take the example of 2-ethylbutanol, Figure 8 shows the process of butaldehyde deprotonation attacking acetaldehyde for nucleophilic addition to complete the aldol reaction. The strong base site, $\text{Ca}[\text{O}^{2-}]\text{Ca}$, enhances the reactivity of the aldol reaction by promoting the α -C deprotonation of butaldehyde. The reaction mechanism between the other aldehydes is similar to that described above. aldol reaction is the step to form C–C, and its reactivity is the key to determining the length of the carbon chain of the product alcohols. In the follow-up study, the yield of higher alcohols (C_{6+}) can be further increased by improving catalyst alkalinity.

Reusability of the Spent Catalyst. The reusability of the optimal HAP4–0.5Ni catalyst was investigated by reusing the spent catalyst in the ethanol upgrading reaction. In order to reduce the error caused by the loss of catalyst, we scaled up the reaction by 50 times, and the results are shown in Figure 9. It is observed that the ethanol conversion has been maintained between 21 and 23%, and the selectivity of butanol and C_{6+} –OH

OH was almost constant. Combined with more than 200 h of stable operation on a fixed bed (Figure 4C), the results clearly indicated the reusability of the HAP4–0.5Ni catalyst.

CONCLUSIONS

This study highlights the importance of the basicity of the Ni/HAP catalysts on the C_{6+} –OH selectivity in the ethanol upgrading reaction. The Ni/HAP catalysts with different densities and strengths of base sites were prepared by controlling the pH of the precipitated solution and were used in ethanol conversion. The results showed that the ethanol reaction rate and the yield of C_{6+} –OH is directly proportional to the amount of $\text{Ni}^{\delta+}$ and the concentration of strong basic $\text{Ca}[\text{O}^{2-}]\text{Ca}$ site, respectively.

The reaction and characterization results showed that Ni loading on HAP4 formed many $\text{Ni}^{\delta+}$ and $\text{Ca}[\text{O}^{2-}]\text{Ca}$ active site pairs, which promoted the continuous condensation of ethanol to higher alcohols. This finding provides insights into the design of a more active and selective catalyst for improving the selectivity of C_{6+} –OH in the future.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.4c03891>.

Materials; catalyst preparation; catalyst characterization methods; theoretical calculations methods; catalytic performance evaluation; N_2 adsorption and desorption isotherms and pore size distribution for the HAPx–0.5Ni; DRIFTS spectra of HAPx–0.5Ni catalysts (4000–3000 cm^{-1}); DRIFTS spectra of HAP4–0.5Ni catalyst before and after reduction; CO_2 -TPD curves of HAP4–0.5Ni catalyst and blank experiment which did not adsorb CO_2 during the test; H_2 -TPR curves of NiO with different masses; fitting curve of the relationship between hydrogen consumption and TPR peak area; H_2 -TPR curves of HAP4; results of ethanol reaction on different catalysts; typical GC chromatogram and main products after ethanol upgrading over HAP4–0.5Ni; and aldol reaction process (PDF)

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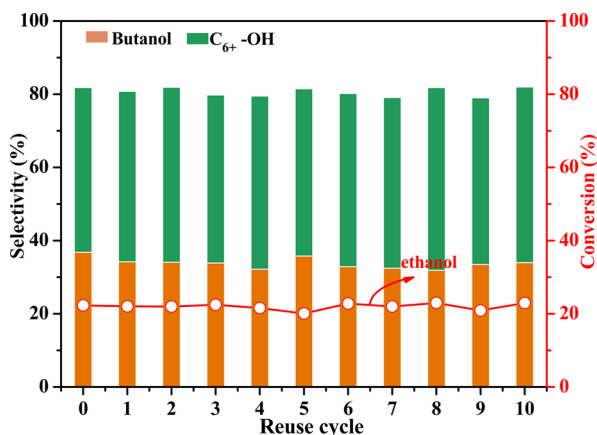


Figure 9. Reusability of the spent HAP4–0.5Ni catalyst for ethanol upgrading. Conditions: 5 g of catalyst, 200 °C for 24 h, 0.1 MPa N_2 , 50 g of ethanol.

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The manuscript was written through the contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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